Exercises 7.3 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 23, 25, 27, 29, 37.

1–4 (a) Determine an appropriate trigonometric substitution. (b) Apply the substitution to transform the integral into a trigonometric integral. Do not evaluate the integral.

$\cos^2 x + \sin^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$\sec^2 x - 1 = \tan^2 x$

1. $\int \frac{x^3}{\sqrt{1+x^2}}\,dx$

let $x = \tan\theta$; $dx = \sec^2 x\,d\theta$

$= \int \frac{\tan^3\theta}{\sqrt{1+\tan^2\theta}} \cdot \sec^2 x\,d\theta$

$= \int \frac{\tan^3\theta}{\sec\theta} \cdot \sec^2\theta\,d\theta$

$= \int \sec\theta\tan^3\theta\,d\theta$

3. $\int \frac{x^2}{\sqrt{x^2-2}}\,dx$

let $x = \sqrt{2}\sec\theta$; $dx = \sqrt{2}\sec\theta\tan\theta\,d\theta$

$= \int \frac{2\sec^2\theta}{\sqrt{2\sec^2\theta - 2}} \cdot \sqrt{2}\sec\theta\tan\theta\,d\theta$

$= \int \frac{2\sec^2\theta}{\sqrt{2}\tan\theta} \sqrt{2}\sec\theta\tan\theta\,d\theta$

$= 2\int \sec^2\theta\,d\theta$

5. $\int \frac{x^3}{\sqrt{1-x^2}}\,dx \qquad x = \sin\theta$

$dx = \cos\theta\, d\theta$

$= \int \frac{\sin^3\theta}{\sqrt{1-\sin^2\theta}} \cos\theta\, d\theta$



$\cos\theta = \sqrt{1-x^2}$

$= \int \sin^3\theta\, d\theta$

$= \int (1-\cos^2\theta)\sin\theta\, d\theta$

Let $u = \cos\theta$; $\frac{du}{d\theta} = -\sin\theta \Rightarrow d\theta = \frac{du}{-\sin\theta}$

$= \int (1-u^2)\sin\theta \cdot \frac{du}{-\sin\theta}$

$= -\int 1-u^2\,du = -\left(u - \frac{u^3}{3}\right) + C$

$= -\left(\cos\theta - \frac{\cos^3\theta}{3}\right) + C$

$= -\cos\theta + \frac{\cos^3\theta}{3} + C$

$= -\sqrt{1-x^2} + \frac{(1-x^2)^{\frac{3}{2}}}{3} + C$

7. $\int \frac{\sqrt{4x^2 - 25}}{x}\,dx \qquad x = \frac{5}{2}\sec\theta$

$dx = \frac{5}{2}\sec\theta\tan\theta\,d\theta$

$= \int \frac{\sqrt{4 \times \frac{25}{4}\sec^2\theta - 25}}{\frac{5}{2}\sec\theta} \cdot \frac{5}{2}\sec\theta\tan\theta\,d\theta$

$= \int \frac{\sqrt{25\sec^2\theta - 25}}{\frac{5}{2}\sec\theta} \cdot \frac{5}{2}\sec\theta\tan\theta\,d\theta$

$= 5\int \tan^2\theta\,d\theta$

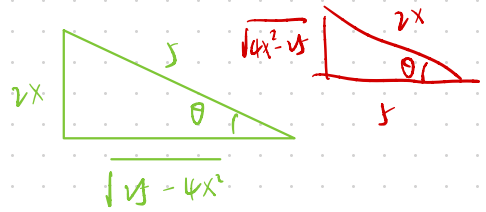
$= 5\int \sec^2\theta - 1\,d\theta$

$= 5\tan\theta - 5\theta + C$

$= \frac{10x}{\sqrt{25 - 4x^2}} - 5\tan^{-1}\left(\frac{2x}{\sqrt{25 - 4x^2}}\right)$

$\frac{1}{\cos\theta} = \frac{2}{5}x$

$\cos\theta = \frac{5}{2x}$



$\tan\theta = \frac{2x}{\sqrt{25 - 4x^2}}$

$\theta = \tan^{-1}\left(\frac{2x}{\sqrt{25 - 4x^2}}\right)$

9. $\int x^3\sqrt{16+x^2}\,dx$

$\cos^2 x + \sin^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

Let $x = 4\tan\theta$; $dx = 4\sec^2\theta\,d\theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$= \int 64\tan^3\theta \sqrt{16+16\tan^2\theta}\cdot 4\sec^2\theta\,d\theta$

$= \int 64\tan^3\theta \cdot 4\sec\theta \cdot 4\sec^2\theta\,d\theta$

$\tan\theta = \frac{x}{4}$

$= 1024\int \sec\theta\tan^3\theta\,d\theta$

$= 1024\int \sec^2\theta(\sec^2\theta - 1)\sec\theta\tan\theta\,d\theta$

Let $u = \sec\theta$; $\frac{du}{d\theta} = \sec\theta\tan\theta \Rightarrow d\theta = \frac{du}{\sec\theta\tan\theta}$

$= 1024\int u^2(u^2-1)\,du$

$= 1024\int u^4 - u^2\,du$

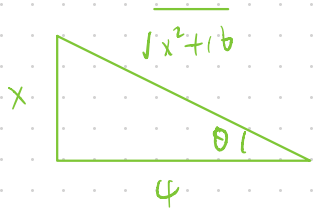
$= 1024\left(\frac{u^5}{5} - \frac{u^3}{3}\right) + C$

$= \frac{1024}{5}u^5 - \frac{1024}{3}u^3 + C$

$= \frac{1024}{5}\sec^5\theta - \frac{1024}{3}\sec^3\theta + C$

$= \frac{1024}{5}\frac{(x^2+16)^{\frac{5}{2}}}{1024} - \frac{1024}{3}\frac{(x^2+16)^{\frac{3}{2}}}{64} + C$

$= \frac{1}{5}(x^2+16)^{\frac{5}{2}} - \frac{16}{3}(x^2+16)^{\frac{3}{2}} + C$



$\cos\theta = \frac{4}{\sqrt{x^2+16}}$

$\sec\theta = \frac{\sqrt{x^2+16}}{4}$

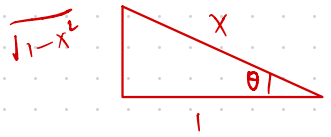
11. $\int \frac{\sqrt{x^2-1}}{x^4}\,dx$

$\sec^2 x - 1 = \tan^2 x$

let $x = \sec\theta$; $dx = \sec\theta\tan\theta\,d\theta$

$= \int \frac{\sqrt{\sec^2\theta - 1}}{\sec^4\theta} \cdot \sec\theta\tan\theta\,d\theta$

$\cos\theta = \frac{1}{x}$



$= \int \frac{\tan^2\theta}{\sec^3\theta}\,d\theta$

$= \int \sin^2\theta\cos\theta\,d\theta$

$\sin\theta = \frac{\sqrt{1-x^2}}{x}$

let $u = \sin\theta$; $\frac{du}{d\theta} = \cos\theta \Rightarrow d\theta = \frac{du}{\cos\theta}$

$= \int u^2 \cdot \cos\theta \cdot \frac{du}{\cos\theta}$

$= \int u^2\,du = \frac{u^3}{3} + C$

$= \frac{\sin^3\theta}{3} + C$

$= \frac{1}{3}\left(\frac{\sqrt{1-x^2}}{x}\right)^3 + C$

13. $\int_0^a \frac{dx}{(a^2+x^2)^{3/2}}, \quad a>0$

$1+\tan^2 x = \sec^2 x$

Let $x = a\tan\theta$; $dx = a\sec^2\theta\, d\theta$

$\theta = \tan^{-1}\left(\frac{x}{a}\right)$

when $x = a$, $\theta = \tan^{-1}\left(\frac{a}{a}\right) = \frac{\pi}{4}$

$x = 0$, $\theta = \tan^{-1}(0) = 0$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{(a^2+a^2\tan^2\theta)^{\frac{3}{2}}} \cdot a\sec^2\theta\, d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{(a^2\sec^2\theta)^{\frac{3}{2}}} \cdot a\sec^2\theta\, d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{a^3\sec^3\theta}\, a\sec^2\theta\, d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{a^2\sec\theta}\, d\theta$$

$$= \frac{1}{a^2}\int_0^{\frac{\pi}{4}} \cos\theta\, d\theta$$

$$= \frac{1}{a^2}\left[\sin\theta\right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{a^2}\left[\frac{\sqrt{2}}{2} - 0\right)$$

$$= \frac{\sqrt{2}-2}{2a^2} \times \frac{\sqrt{2}}{2a^2} = \frac{1}{\sqrt{2}a^2}$$

15. $\int_2^3 \frac{dx}{(x^2-1)^{3/2}}$ $\qquad \sec^2\theta - 1 = \tan^2\theta$

let $x = \sec\theta$; $dx = \sec\theta\tan\theta\, d\theta$

$\theta = \cos^{-1}\left(\frac{1}{x}\right)$

when $x = 3$, $\theta = \cos^{-1}\left(\frac{1}{3}\right) = 1.23$

$x = 2$, $\theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$

$$= \int_{\frac{\pi}{3}}^{1.23} \frac{1}{(\sec^2\theta - 1)^{\frac{3}{2}}} \cdot \sec\theta\tan\theta\, d\theta$$

$$= \int_{\frac{\pi}{3}}^{1.23} \frac{1}{\tan^3\theta} \cdot \sec\theta\tan\theta\, d\theta$$

$$= \int_{\frac{\pi}{3}}^{1.23} \frac{\sec\theta}{\tan^2\theta}\, d\theta$$

$$= \int_{\frac{\pi}{3}}^{1.23} \frac{\cos\theta}{\sin^2\theta}\, d\theta$$

let $u = \sin\theta$; $\frac{du}{d\theta} = \cos\theta \Rightarrow d\theta = \frac{du}{\cos\theta}$

when $\theta = 1.23$, $u = \sin(1.23) = 0.94$

$\theta = \frac{\pi}{3}$, $u = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

$$= \int_{\frac{\sqrt{3}}{2}}^{0.94} \frac{1}{u^2}\, du$$

$$= -\left[u^{-1}\right]_{\frac{\sqrt{3}}{2}}^{0.94}$$

$$= -\left[0.94^{-1} - \left(\frac{\sqrt{3}}{2}\right)^{-1}\right] = +0.091$$

17. $\int_0^{1/2} x\sqrt{1-4x^2}\,dx$

$1-\sin^2\theta = \cos^2\theta$

let $2x = \cos\theta$

$x = \frac{\cos\theta}{2}$, $dx = -\frac{\sin\theta}{2}\,d\theta$

$\theta = \cos^{-1}(2x)$

when $x = \frac{1}{2}$, $\theta = \cos^{-1}(2\times\frac{1}{2}) = 0$

$x = 0$, $\theta = \cos^{-1}(0) = \frac{\pi}{2}$

$= \int_{\frac{\pi}{2}}^{0} \frac{\cos\theta}{2} \cdot \sqrt{1-\cos^2\theta} \cdot \left(-\frac{\sin\theta}{2}\right) d\theta$

$= -\frac{1}{4}\int_{\frac{\pi}{2}}^{0} \cos\theta \cdot \sin^2\theta\, d\theta$

let $u = \sin\theta$; $\frac{du}{d\theta} = \cos\theta \Rightarrow d\theta = \frac{1}{\cos\theta}\,du$

when $\theta = 0$, $u = \sin 0 = 0$

$\theta = \frac{\pi}{2}$, $u = \sin\frac{\pi}{2} = 1$

$= \frac{1}{4}\int_0^1 u^2 \cdot du$

$= \frac{1}{4} \times \frac{1}{3}[u^3]_0^1$

$= \frac{1}{12}(1-0) = \frac{1}{12}$

19. $\int \frac{\sqrt{x^2-9}}{x^3}\,dx$

$\sec^2\theta - 1 = \tan^2\theta$

Let $x = 3\sec\theta$; $dx = 3\sec\theta\tan\theta\,d\theta$

$= \int \frac{\sqrt{9\sec^2\theta - 9}}{27\sec^3\theta} \cdot 3\sec\theta\tan\theta\,d\theta$

$= \int \frac{3\tan\theta}{27\sec^3\theta} \cdot 3\sec\theta\tan\theta\,d\theta$

$= \frac{1}{3}\int \frac{\tan^2\theta}{\sec^2\theta}\,d\theta$

$= \frac{1}{3}\int \sin^2\theta\,d\theta$

$\frac{1}{3}\cdot\frac{1}{2}\int 1-\cos 2\theta\,d\theta$

$= \frac{1}{6}\left(\theta - \frac{\sin 2\theta}{2}\right) + C$

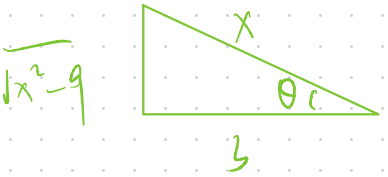
$= \frac{1}{6}\theta - \frac{\sin\theta\cos\theta}{6} + C$

$= \frac{1}{2}\cos^{-1}\left(\frac{3}{x}\right) - \frac{\frac{3\sqrt{x^2-9}}{x^2}}{2} + C$

$\frac{1}{6}\cdot\frac{\sqrt{x^2-9}}{x}\cdot\frac{3}{x} = \frac{\sqrt{x^2-9}}{2x^2}$

$\frac{1}{\cos\theta} = \frac{x}{3}$

$\cos\theta = \frac{3}{x}$



$\sqrt{x^2-9}$

$\theta = \cos^{-1}\left(\frac{3}{x}\right)$

$\sin\theta = \frac{\sqrt{x^2-9}}{x}$

23. $\int \frac{x}{\sqrt{x^2-7}}\,dx$

$\sec^2 x - 1 = \tan^2 x$

let $x = \sqrt{7}\sec\theta$; $dx = \sqrt{7}\sec\theta\tan\theta\,d\theta$

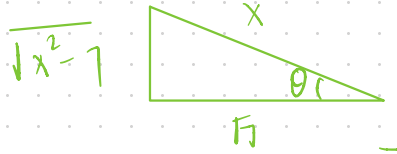
$$= \int \frac{\sqrt{7}\sec\theta}{\sqrt{7\sec^2\theta - 7}} \cdot \sqrt{7}\sec\theta\tan\theta\,d\theta$$

$$= \int \frac{\sqrt{7}\sec\theta}{\sqrt{7}\tan\theta} \cdot \sqrt{7}\sec\theta\tan\theta\,d\theta$$

$$= \sqrt{7}\int \sec^2\theta\,d\theta$$

$$= \sqrt{7}\tan\theta + C$$

$$= \sqrt{x^2-7} + C$$



$\tan\theta = \frac{\sqrt{x^2-7}}{\sqrt{7}}$

25. $\int \frac{\sqrt{1+x^2}}{x}\,dx$

$1 + \tan^2 x = \sec^2 x$

$\sec^2 x =$

$\csc x \quad -\csc x \cot x$

$\cot x \quad -\csc^2 x$

Let $x = \tan\theta$; $dx = \sec^2\theta\, d\theta$

$= \int \frac{\sqrt{1+\tan^2\theta}}{\tan\theta} \cdot \sec^2\theta\, d\theta$

$= \int \frac{\sec\theta}{\tan\theta} \sec^2\theta\, d\theta$ $\quad (1+\tan^2\theta)$

$= \int \frac{\sec\theta(1+\tan^2\theta)}{\tan\theta}\, d\theta$

$= \int \frac{\sec^2\theta}{\tan\theta}\, d\theta$

$= \int \frac{1}{\sin\theta} + \sec\theta\tan\theta\, d\theta$

$= -\ln|\csc x + \cot x| + \sec\theta + C$

$\int \csc x\, dx$

$= \int \csc x \cdot \frac{\csc x + \cot x}{\csc x + \cot x}\, dx$

$= \int \frac{\csc^2 x + \csc x \cot x}{\csc x + \cot x}\, dx$

let $u = \csc x + \cot x$; $\frac{du}{dx} = -\csc\cot x - \csc^2 x$

$dx = \frac{du}{-(\csc\cot x + \csc^2 x)}$

$= -\int \frac{1}{u}\, du = -\ln|\csc x + \cot x| + C = \ln|(\csc x + \cot x)^{-1}| + C$

$= \ln|\csc x - \cot x| + C$

27. $\int_0^{0.6} \frac{x^2}{\sqrt{9-25x^2}}\,dx$

Let $5x = 3\sin\theta$

$x = \frac{3}{5}\sin\theta$; $dx = \frac{3}{5}\cos\theta\,d\theta$

$\theta = \sin^{-1}\left(\frac{5x}{3}\right)$

when $x = 0.6$, $\theta = \sin^{-1}\left(\frac{3}{3}\right) = \frac{\pi}{2}$

$x = 0$, $\theta = \sin^{-1}(0) = 0$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{9}{25}\sin^2\theta}{\sqrt{9-9\sin^2\theta}} \cdot \frac{3}{5}\cos\theta\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{9}{125}\sin^2\theta\,d\theta$$

$$= \frac{9}{125}\int_0^{\frac{\pi}{2}} 1-\cos^2\theta\,d\theta$$

$$= \frac{9}{125}\int_0^{\frac{\pi}{2}} 1-\frac{\cos 2\theta+1}{2}\,d\theta$$

$$= \frac{9}{125}[\theta]_0^{\frac{\pi}{2}} - \frac{9}{125}\times\frac{1}{2}\int_0^{\frac{\pi}{2}} \cos 2\theta + 1\,d\theta$$

$$= \frac{9\pi}{250} - \frac{9}{250}\left[\frac{\sin 2\theta}{2} + \theta\right]_0^{\frac{\pi}{2}}$$

$$= \frac{9\pi}{250} - \frac{9}{250}\left[\frac{\sin\pi}{2} + \frac{\pi}{2} - \left(\frac{0}{2}+0\right)\right]$$

$$= \frac{9\pi}{250} - \frac{9}{250}\times\frac{\pi}{2}$$

$$= \frac{9\pi}{500}$$

29. $\int \frac{dx}{\sqrt{x^2 + 2x + 5}}$

$= \int \frac{dx}{\sqrt{(x+1)^2 + 4}}$

let $x+1 = 2\tan\theta$;

$x = 2\tan\theta - 1$; $dx = 2\sec^2\theta\, d\theta$

$= \int \frac{1}{\sqrt{4\tan^2\theta + 4}} \cdot 2\sec^2\theta\, d\theta$

$= \int \frac{1}{2\sec\theta} \, 2\sec^2\theta\, d\theta$

$= \int \sec\theta\, d\theta$

$= \ln|\sec\theta + \tan\theta| + C$